

2.4 Differentiation

Pure Mathematics 2 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Implicit differentiation differentiates y as $y(x)$, introducing dy/dx . Parametric curves use $dy/dx = (dy/dt)/(dx/dt)$; chain and product rules handle composite products. Set $dy/dx = 0$ carefully because a zero numerator with zero denominator may need separate analysis.

Worked example: follow the reasoning

On $x^2 + y^2 = 25$, $dy/dx = -x/y$; at $(3,4)$ gradient $-3/4$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 For $x = t^2 + 1$, $y = t^3$, calculate dy/dx at $t = 2$.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

$\frac{dy}{dt}=3t^2$, $\frac{dx}{dt}=2t$ [1]; $\frac{dy}{dx}=\frac{3t}{2}$ [1]; at 2 gradient 3 [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.